Write a Python Program using Perceptron

Neural Network to

recognize even and

odd numbers.Given numbers are in ASCII form 0 to 9.

In [1]:
```
import numpy as np
```

In [2]:
```
X=[[int(b) for b in format(ord(str(d)),'08b')] for d in range(10)]
labels=[1 if d%2==0 else 0 for d in range(10)]
```

In [3]:
```
def step(x):
    return np.where(x>=0,1,0)
```

In [4]:
```
w=np.zeros(8)
lr=0.1
b=0.1
```

In [5]:
```
while True:
    err_cnt = 0
    for x, target in zip(X, labels):
        net = np.dot(x, w) + b
        output = step(net)
        err = target - output
        if err != 0:
            w += lr * err * np.array(x)
            b += lr * err
            err_cnt += 1
    if err_cnt == 0:
        break
```

In [6]:
```
num = int(input("Enter any number 0-9: "))
test = [int(bit) for bit in format(ord(str(num)), '08b')]
net = np.dot(test, w) + b
output = step(net)
if output == 1:
    print(num, "is Even")
else:
    print(num, "is Odd")
```

```
5 is Odd
```

In [ ]:
```
```